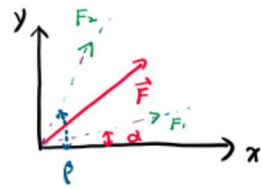


Topic 1 – Force Analysis

(1) Resultant Forces



$$|F| \cdot \cos(\theta_F) = |F_1| \cos(\alpha) + |F_2| \cos(\beta)$$

$$|F| \cdot \sin(\theta_F) = |F_1| \sin(\alpha) + |F_2| \sin(\beta)$$

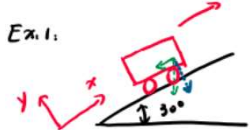
(2) Robot Driving Forces

$$f_{driving} = \frac{\tau \text{ (Torque)}}{R \text{ (Wheel Radius)}}$$

(3) Driven Wheel

$$f_{resistance} = mg \times C_r(v) \text{ (Rolling Resistance)}$$

(4) Robot on a Slope



Upward:

$$F_{net} = \frac{\tau}{R} - mg \sin(30) - mg \cos(30) \times C_r(v)$$

Downward:

$$F_{net} = \frac{\tau}{R} + mg \sin(30) - mg \cos(30) \times C_r(v)$$

Topic 2 – Navigation

(5) Global to Local Coord Transformation

$$\begin{bmatrix} x'_A \\ y'_A \end{bmatrix} = \begin{bmatrix} x'_o \\ y'_o \end{bmatrix} + \begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix} \times \begin{bmatrix} x_A \\ y_A \end{bmatrix}$$

(6) Local to Global Coord Transformation

$$\begin{bmatrix} x_A \\ y_A \end{bmatrix} = \begin{bmatrix} \cos \theta & \sin \theta \\ -\sin \theta & \cos \theta \end{bmatrix} \cdot \left(\begin{bmatrix} x'_A \\ y'_A \end{bmatrix} - \begin{bmatrix} x'_o \\ y'_o \end{bmatrix} \right)$$

$$\text{dist}_{A \rightarrow B} = \sqrt{(x'_A - x'_B)^2 + (y'_A - y'_B)^2}$$

(7) CT for Velocity Vector (Local to Global)

$$\dot{x}' = \dot{x}'_o - \sin(\theta) \cdot \dot{\theta} \cdot x + \cos(\theta) \cdot \dot{x} - \cos \theta \cdot \dot{\theta} \cdot y - \sin \theta \cdot \dot{y}$$

$$\dot{y}' = \dot{y}'_o + \cos(\theta) \cdot \dot{\theta} \cdot x + \sin(\theta) \cdot \dot{x} - \sin(\theta) \cdot \dot{\theta} \cdot y + \cos(\theta) \cdot \dot{y}$$

Variable	Meaning
$\dot{x}, \dot{y}$	Velocity of object in local frame
$\dot{x}', \dot{y}'$	Velocity of object in global frame
$\dot{x}'_o, \dot{y}'_o$	Velocity of robot origin in global frame
x, y	Position of object in local frame
θ	Robot's heading angle (rotation from global x-axis)
$\dot{\theta}$	Angular velocity of the robot (how fast it's turning)
R	Rotation matrix $\begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix}$

(8) CT Velocity (linear motion/slow rotate)

$$\text{if } \dot{\theta} \approx 0: \begin{bmatrix} \dot{x}' \\ \dot{y}' \end{bmatrix} = R \times \begin{bmatrix} \dot{x} \\ \dot{y} \end{bmatrix}$$

(9) Robot Control CMD (No Rot. & Rot.)

$$x' = x_{initial} + \text{disp } x'_{1 \rightarrow 2} + \text{disp } x'_{2 \rightarrow 3}$$

$$y' = y_{initial} + \text{disp } y'_{1 \rightarrow 2} + \text{disp } y'_{2 \rightarrow 3}$$

$$\begin{bmatrix} x' \\ y' \end{bmatrix} = \begin{bmatrix} x'_o \\ y'_o \end{bmatrix} + \begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix} \cdot \begin{bmatrix} \Delta x \\ \Delta y \end{bmatrix}$$

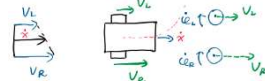
(10) Wheel Motion, $\dot{x} = IV$ of Heading

◆ Case 1: Straight Motion

$$\dot{\phi}_L = \dot{\phi}_R \Rightarrow \dot{x} = R \cdot \dot{\phi}_L = R \cdot \dot{\phi}_R$$

◆ Case 2: General Case

$$\dot{x} = \frac{V_L + V_R}{2} = \frac{R \cdot \dot{\phi}_L + R \cdot \dot{\phi}_R}{2}$$



◆ Case 3: Rotation In Place

$$\dot{\phi}_L = -\dot{\phi}_R \Rightarrow \dot{x} = \frac{R \cdot \dot{\phi}_L + R \cdot \dot{\phi}_R}{2} = 0$$

(11) FWD Linear & Angular Velocity (Local)

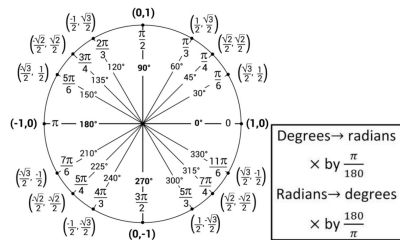
$$\dot{x} = \frac{R}{2} (\dot{\phi}_L + \dot{\phi}_R)$$

$$\dot{\theta} = \frac{R}{2L} (\dot{\phi}_R - \dot{\phi}_L)$$

(12) Instantaneous Radius of Rotation

$$\text{Turning Left: } D = L \times \frac{\dot{\phi}_L + \dot{\phi}_R}{\dot{\phi}_R - \dot{\phi}_L}$$

$$\text{Turning Right: } D = L \times \frac{\dot{\phi}_L + \dot{\phi}_R}{\dot{\phi}_L - \dot{\phi}_R}$$



(13) Trajectory Case 1 – Linear [*11]

Step 1: From A to B, $\dot{\phi}_L = \dot{\phi}_R$

$$\dot{x} = \frac{R}{2} \cdot (\dot{\phi}_L + \dot{\phi}_R) = (0.02) \cdot (\dot{\phi}_L + \dot{\phi}_R)$$

$$\Rightarrow \dot{x} = \frac{10m}{0.02} = \frac{10m}{0.02m} = 1 \text{ m/s}$$

$$\Rightarrow \dot{\phi}_L = \dot{\phi}_R = \frac{1}{2} = 25 \text{ rad/sec}$$

Step 2: 90° turn (Counter Clock-Wise at B)

$$\dot{\theta} = \frac{\dot{\phi}_R - \dot{\phi}_L}{2} = \frac{\pi}{2} = \frac{\pi}{2} \text{ rad/sec}$$

$$\dot{x} = 0 \text{ m/s}, \quad \dot{\theta} = \frac{90^\circ}{\frac{\pi}{2}} = \frac{90^\circ}{1.57} = 90 \text{ %/s}$$

$$\Rightarrow \left(\dot{\theta} = \frac{\pi}{2} = \frac{180^\circ}{2} = 90^\circ \right) \cdot \dot{\phi}_L = \dot{\phi}_R$$

$$\Rightarrow \dot{\phi}_L = \dot{\phi}_R = \frac{90^\circ}{2} = 45 \text{ %/s}$$

$$\Rightarrow \dot{\phi}_L = \dot{\phi}_R = \frac{90^\circ}{2} = 45 \text{ %/s}$$

$$\Rightarrow \dot{\phi}_L = \dot{\phi}_R = \frac{90^\circ}{2} = 45 \text{ %/s}$$

(14) Trajectory Case 2 – Arc Trajectory

Solution:

Step 1: Determine $\dot{\theta}_{A \rightarrow B}$

$$\dot{\theta}_{A \rightarrow B} = \frac{\theta_B - \theta_A}{t} = \frac{30^\circ - 0^\circ}{10} = 0.314 \text{ rad/s}$$

Step 2: Determine $\dot{x}$

$$\dot{x} \cdot dt = \text{length of arc}$$

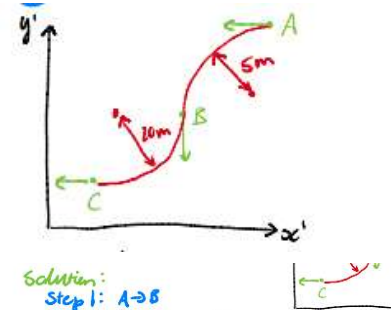
$$\Rightarrow \dot{x} = \frac{\text{length}}{t} = \frac{\pi/10}{10} = \pi \text{ m/s}$$

Step 3: Replace $\dot{x}$ and $\dot{\theta}$ with calculated values

$$\left\{ \begin{array}{l} \pi = \frac{\pi}{2} \cdot (\dot{\phi}_L - \dot{\phi}_R) \quad ① \\ 0.314 = \frac{\pi}{2} \cdot (\dot{\phi}_L - \dot{\phi}_R) \quad ② \end{array} \right.$$

$$\left\{ \begin{array}{l} \dot{\phi}_L = 76.9 \text{ rad/sec} \\ \dot{\phi}_R = 20.1 \text{ rad/sec} \end{array} \right.$$

(15) Trajectory Case 3 – General Path



Solution:

Step 1: A to B

$$\dot{\theta}_{A \rightarrow B} = \frac{\theta_B - \theta_A}{t} = \frac{30^\circ - 0^\circ}{10} = \frac{\pi}{20} \text{ rad/s}$$

$$\dot{x}_{A \rightarrow B} = \frac{\text{length of arc A to B}}{10} = \frac{30^\circ \cdot 5}{10} = \frac{\pi}{4} \text{ m/s}$$

$$\left\{ \begin{array}{l} \dot{x}_{A \rightarrow B} = \frac{\pi}{4} = \frac{0.01}{2} \cdot (\dot{\phi}_L + \dot{\phi}_R) \quad ① \\ \dot{\theta}_{A \rightarrow B} = \frac{\pi}{20} = \frac{0.01}{2} \cdot (\dot{\phi}_L - \dot{\phi}_R) \quad ② \end{array} \right.$$

Solve $\dot{\phi}_L$ and $\dot{\phi}_R$ using equations 1 and 2

Step 2: B to C

$$\dot{\theta}_{B \rightarrow C} = \frac{\theta_C - \theta_B}{t} = \frac{\pi - 30^\circ}{10} = -\frac{\pi}{20} \text{ rad/s}$$

$$\dot{x}_{B \rightarrow C} = \frac{\text{length of arc B to C}}{10} = \frac{1/4 \cdot \pi \cdot 20}{10} = \pi \text{ m/s}$$

$$\left\{ \begin{array}{l} \dot{x}_{B \rightarrow C} = \pi = \frac{0.01}{2} \cdot (\dot{\phi}_L + \dot{\phi}_R) \\ \dot{\theta}_{B \rightarrow C} = \frac{\pi}{20} = \frac{0.01}{2} \cdot (\dot{\phi}_L - \dot{\phi}_R) \end{array} \right.$$

(16) Trajectory Case 4 – Math Function

Given $y' = \sqrt[3]{x}$ from $x'=1$ to $x'=1.5$ m:

$$\theta_{\text{final}(2)} = \arctan\left(\frac{d}{dx}(y')\right)_{x=1.5} \text{ (degrees, convert to rad after)}$$

$$\theta_{\text{initial}(1)} = \arctan\left(\frac{d}{dx}(y')\right)_{x=1} \text{ (degrees, convert to rad after)}$$

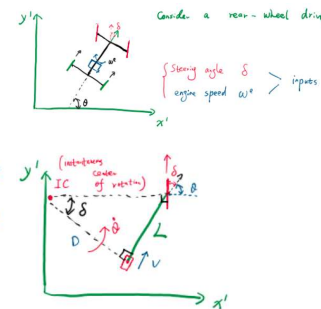
$$\dot{\theta} = \frac{\theta_{\text{final}} - \theta_{\text{initial}}}{\text{time}} \text{ (rad/s)}$$

$$\text{distance} = \frac{\sqrt{(y_2 - y_1)^2 + (x_2 - x_1)^2}}{2 \sin\left(\frac{|\theta_2 - \theta_1|}{2}\right)} \text{ (m)}$$

$$\dot{x} = \frac{\text{Distance} \times |\theta_2 - \theta_1|}{t} \text{ (m/s)}$$

Find Phi Dot using [11]

(17) RWD Ackermann Steering



◆ Rear-Wheel Drive (RWD) — Drive in the Rear

Step 1: Compute linear velocity

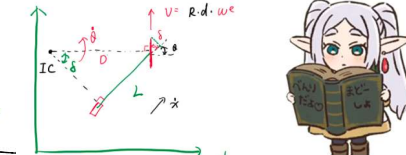
$$\dot{x} = v = R \cdot d \cdot \omega'$$

Step 2: Use turning radius $D = \frac{L}{\tan \delta}$

Step 3: Compute angular velocity

$$\dot{\theta} = \frac{v}{D} = \frac{R \cdot d \cdot \omega' \cdot \tan \delta}{L}$$

(18) FWD Ackermann Steering



◆ Front-Wheel Drive (FWD) — Drive in the Front

Step 1: Use geometry: $\frac{1}{D} = \sin \delta \Rightarrow D = \frac{L}{\sin \delta}$

Step 2: Compute angular velocity

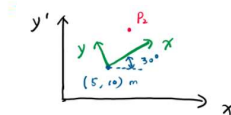
$$\dot{\theta} = \frac{v}{D} = \frac{R \cdot d \cdot \omega' \cdot \sin \delta}{L}$$

Step 3: Resolve forward velocity along heading:

$$\dot{x} = v \cdot \cos \delta = R \cdot d \cdot \omega' \cdot \cos \delta$$

Topic 3 – Sensing with LIDAR

(19) Distance Sensing with Lidar



Reading from the LIDAR

The position of P2: (10m, 45deg)

The position of LIDAR on the scuttle robot: [PoP1] = 254mm

Po = Position of the center of the wheel base. Origin of the local coordinate

Q: What is the position of P2 in the global coordinate?

Step 1: Find P2 in the local coordinate

$$x_{P2} = d_i \cos \phi + [P_0 P_1] = 10 \cos(45) + 0.254 = 7.254 \text{ m}$$

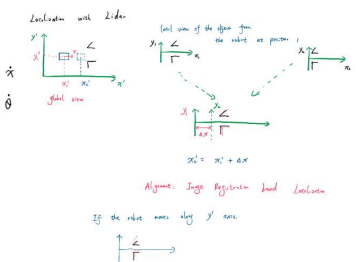
$$y_{P2} = d_i \sin \phi = 10 \sin(45) = 7 \text{ m}$$

Step 2: Find P2 in the global coordinate

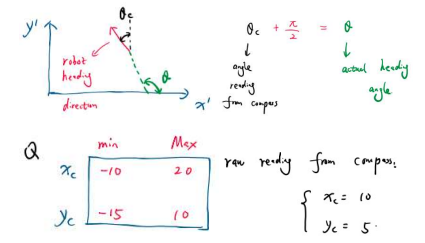
Use Equation #9

$$\begin{bmatrix} x'_{P2} \\ y'_{P2} \end{bmatrix} = \begin{bmatrix} 5 \\ 10 \end{bmatrix} + \begin{bmatrix} \cos(30) & -\sin(30) \\ \sin(30) & \cos(30) \end{bmatrix} \times \begin{bmatrix} 7.254 \\ 7 \end{bmatrix} = \begin{bmatrix} 7.78 \text{ m} \\ 19.69 \text{ m} \end{bmatrix}$$

(20) Localization with Lidar



(21) Compass Calibration



$$x_{c, \text{scale}} = \frac{2 \times (x_c - x_{c, \text{min}})}{x_{c, \text{max}} - x_{c, \text{min}}} - 1$$

$$y_{c, \text{scale}} = \frac{2 \times (y_c - y_{c, \text{min}})}{y_{c, \text{max}} - y_{c, \text{min}}} - 1$$

$$\text{Heading angle: } \theta_c = \arctan\left(\frac{x_{c, \text{scale}}}{y_{c, \text{scale}}}\right) = \arctan\left(\frac{1/3}{3/5}\right) = 0.5$$

$$\theta = \theta_c + \frac{\pi}{2}$$

Topic 4 – Controls

(22) Motor Dynamics Equations

$$L \cdot \dot{I}_L + R \cdot I_L = V_i$$

$$J \cdot \dot{\omega} + D \cdot \omega = K_t \cdot I_L - T_L$$

(23) Steady State Assumption

$$\dot{I}_L = 0, \quad \dot{\omega} = 0$$

$$I_L = \frac{V_i}{R}$$

$$\omega = \frac{K_t}{DR} V_i - \frac{T_L}{D}$$

(24) Feedforward Control (Open-Loop)

$$V_i^{\text{FF}} = \frac{DR}{K_t} \left(\omega^d + \frac{T_L^N}{D} \right)$$

(25) Feedback Control (Closed-Loop)

$$E = \omega^d - \omega$$

$$V_i^{\text{FB}} = K_p \cdot E$$

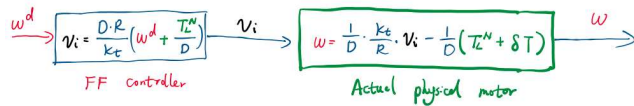
$$V_i = V_i^{\text{FF}} + V_i^{\text{FB}}$$

$$\omega = \omega^d - \frac{\delta T}{D + \frac{K_p K_t}{R}}$$

Symbol	Meaning
L	Motor inductance
R	Electrical resistance
J	Rotational inertia
D	Damping coefficient (friction)
K_t	Motor torque constant
T_L	Load torque
T_L^N	Nominal load torque (used in FF design)
δT	Load disturbance (uncertainty)
K_p	Proportional gain
ω^d	Desired speed
ω	Actual motor speed

(26) Feed Forward

T_L^N + δT
 $\downarrow$
 calibrated nominal value used for controller design
 $\rightarrow$ uncertainty / perturbation on T_L in the working environment.



with the uncertainty δT , the real motor speed using the feedforward controller is:

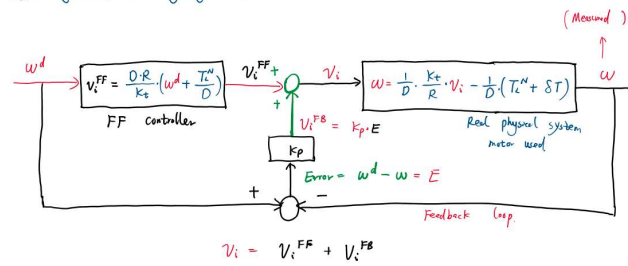
$$\begin{aligned}
 \omega &= \frac{1}{D} \cdot \frac{k_t}{R} \cdot V_i - \frac{1}{D} \cdot (T_L^N + \delta T) \\
 &\downarrow \\
 &\frac{D \cdot R}{k_t} \cdot \left(\omega^d + \frac{T_L^N}{D} \right) \\
 &= \frac{1}{D} \cdot \frac{k_t}{R} \cdot \left[\frac{D \cdot R}{k_t} \cdot \left(\omega^d + \frac{T_L^N}{D} \right) \right] - \frac{1}{D} (T_L^N + \delta T) \\
 &= \omega^d + \frac{T_L^N}{D} - \frac{T_L^N}{D} - \frac{\delta T}{D}
 \end{aligned}$$

$$\Rightarrow \omega = \omega^d - \frac{\delta T}{D}$$

$\downarrow$
desired motor speed

Feedback + Feedforward

$\Rightarrow$ feedback + feedforward.



$$\omega = \omega^d - \frac{\delta T}{D + \frac{k_p \cdot k_t}{R}}$$

$$\begin{aligned}
 \omega &= \frac{k_t}{D \cdot R} \cdot V_i - \frac{1}{D} \cdot (T_L^N + \delta T) \\
 &\downarrow \\
 &\frac{D \cdot R}{k_t} \cdot \left(\omega^d + \frac{T_L^N}{D} \right) + k_p \cdot (\omega^d - \omega)
 \end{aligned}$$

$$\begin{aligned}
 \Rightarrow \omega &= \frac{k_t}{D \cdot R} \cdot \left[\frac{D \cdot R}{k_t} \cdot \left(\omega^d + \frac{T_L^N}{D} \right) + k_p \cdot (\omega^d - \omega) \right] - \frac{1}{D} \cdot (T_L^N + \delta T) \\
 &= \omega^d + \frac{T_L^N}{D} + \frac{k_t \cdot k_p}{D \cdot R} \cdot (\omega^d - \omega) - \frac{1}{D} (T_L^N + \delta T) \\
 \Rightarrow \left[1 + \frac{k_t \cdot k_p}{D \cdot R} \right] \cdot \omega &= \left[1 + \frac{k_t \cdot k_p}{D \cdot R} \right] \cdot \omega^d - \frac{\delta T}{D}
 \end{aligned}$$

$$\Rightarrow \omega = \omega^d - \frac{1}{D + \frac{k_p \cdot k_t}{R}} \cdot \delta T$$

$\downarrow$ Actual motor speed / wheel speed
 $\downarrow$ Desired / target motor speed
 $\downarrow$ reduce the impact of the uncertainty δT on the control performance by choosing a relatively large K_p
 $\downarrow$ uncertainty / perturbation

①. Feedforward (open loop)

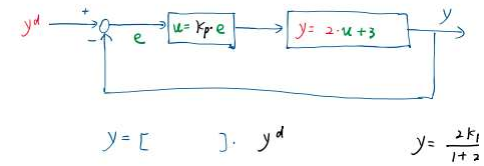
Pros: Directly generate the required control input at steady state; Reduce the settling time.
 Cons: sensitive to uncertainty and disturbance.

②. Feedback controller

Pros: reduce the impact of uncertainty.
 Cons: Take time to learn the steady state control input (longer transition time)

③. Feedforward + Feedback.

Example:



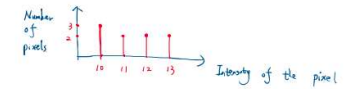
$$\begin{aligned}
 y &= [\quad] \cdot y^d \quad y = \frac{2k_p}{1+2k_p} \\
 \text{Solution:} \quad y &= 2 \cdot u + 3 \\
 &\downarrow \\
 &K_p \cdot e \\
 &= 2 \cdot K_p \cdot e + 3 \\
 &\downarrow \\
 &y^d - y \\
 &= 2 \cdot K_p \cdot (y^d - y) + 3 \\
 \Rightarrow (1 + 2K_p) \cdot y &= 2 \cdot K_p \cdot y^d + 3 \\
 \Rightarrow y &= \frac{2K_p}{1+2K_p} \cdot y^d + \frac{3}{1+2K_p}
 \end{aligned}$$

Example [contrast enhancement for Computer Vision]

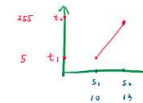
A.

10	11	10
12	13	11
12	13	10

① Draw the histogram of the image



② Determine the function $g(s)$ that can stretch the intensity spectrum to $[5, 255]$



$$\begin{aligned}
 g(s) &= (s - s_1) \cdot \frac{s_2 - s_1}{s_2 - s_1} + s_1 \\
 &= (s - 10) \cdot \frac{255 - 5}{13 - 10} + 5 \\
 &= (s - 10) \cdot \frac{250}{3} + 5
 \end{aligned}$$

A.

10	11	10
12	13	11
12	13	10

$\Rightarrow$

New image B.

5	88	5
172	255	88
172	255	5

